

Solid Geometry & Similarity

ANSWER KEY



Volume of prisms and cylinders

- 1 Find the volume of a rectangular prism with length 8, width 5, and height 6.
 $V = lwh = 8 \times 5 \times 6 = 240$ cubic units.
 - 2 Find the volume of a cylinder with radius 5 cm and height 14 cm, in terms of π .
 $V = \pi r^2 h = \pi(25)(14) = 350\pi$ cm³.
 - 3 A triangular prism has a triangular cross-section of area 24 cm² and length 15 cm. Find its volume.
 $V = (\text{cross-section area}) \times \text{length} = 24 \times 15 = 360$ cm³.
 - 4 Find the volume of a cube with side length 9.
 $V = 9^3 = 729$ cubic units.
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Volume of pyramids and cones

- 5 Find the volume of a square pyramid with base side 10 and height 12.
 $V = \frac{1}{3}(\text{base area})(\text{height}) = \frac{1}{3}(100)(12) = 400$ cubic units.
 - 6 Find the volume of a cone with radius 6 and height 15, in terms of π .
 $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(15) = 180\pi$ cubic units.
 - 7 A cone has volume 300π cubic units and height 25. Find its radius.
 $300\pi = \frac{1}{3}\pi r^2(25) \Rightarrow r^2 = \frac{900}{25} = 36 \Rightarrow r = 6$.
 - 8 A rectangular pyramid has a base 6 by 8 and height 9. Find its volume.
 $V = \frac{1}{3}(6 \times 8)(9) = \frac{1}{3}(48)(9) = 144$ cubic units.
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Volume of spheres

- 9 Find the volume of a sphere with radius 9, in terms of π .
 $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(729) = 972\pi$ cubic units.
- 10 A hemisphere has radius 6. Find its volume, in terms of π .
 $V = \frac{1}{2}\left(\frac{4}{3}\pi r^3\right) = \frac{2}{3}\pi(216) = 144\pi$ cubic units.
- 11 A spherical tank has diameter 10. Find its volume, in terms of π .
Radius = 5. $V = \frac{4}{3}\pi(125) = \frac{500\pi}{3}$ cubic units.

- 12 Find the volume and surface area of a sphere with radius 3, in terms of π .

$$V = \frac{4}{3}\pi(27) = 36\pi \text{ cubic units. } SA = 4\pi(9) = 36\pi \text{ square units.}$$

Surface area of solids

- 13 Find the surface area of a cylinder with radius 4 and height 10, in terms of π .

$$SA = 2\pi r^2 + 2\pi rh = 2\pi(16) + 2\pi(4)(10) = 32\pi + 80\pi = 112\pi.$$

- 14 Find the surface area of a cone with radius 5 and slant height 13, in terms of π .

$$SA = \pi r^2 + \pi rl = 25\pi + 65\pi = 90\pi.$$

- 15 Find the surface area of a sphere with radius 7, in terms of π .

$$SA = 4\pi r^2 = 4\pi(49) = 196\pi.$$

- 16 Find the surface area of a cube with side length 7.

$$SA = 6(7^2) = 6(49) = 294 \text{ square units.}$$

Composite solids

- 17 A solid consists of a cylinder (radius 3, height 10) topped by a cone with the same radius and height 4. Find the total volume, in terms of π .

$$\text{Cylinder: } \pi(9)(10) = 90\pi. \text{ Cone: } \frac{1}{3}\pi(9)(4) = 12\pi. \text{ Total} = 90\pi + 12\pi = 102\pi.$$

- 18 A rectangular box measuring 8 by 6 by 4 has a cylindrical hole (radius 2, drilled fully through the height of 4) removed. Find the remaining volume, in terms of π .

$$\text{Box volume} = 8 \times 6 \times 4 = 192. \text{ Cylinder removed} = \pi(4)(4) = 16\pi. \text{ Remaining volume} = 192 - 16\pi.$$

- 19 A solid consists of a cylinder (radius 4, height 8) topped by a hemisphere of the same radius. Find the total volume, in terms of π .

$$\text{Cylinder: } \pi(16)(8) = 128\pi. \text{ Hemisphere: } \frac{2}{3}\pi(64) = \frac{128\pi}{3}. \text{ Total} = 128\pi + \frac{128\pi}{3} = \frac{512\pi}{3}.$$

- 20 A solid is a rectangular box (10 by 6 by 5) with a square pyramid (base 6 by 6, height 4) carved out of the top. Find the remaining volume.

$$\text{Box: } 10 \times 6 \times 5 = 300. \text{ Pyramid: } \frac{1}{3}(36)(4) = 48. \text{ Remaining volume} = 300 - 48 = 252 \text{ cubic units.}$$

Similar solids

- 21 Two similar cylinders have heights 6 and 9. Find the ratio of their volumes.

$$\text{Scale factor } k = \frac{9}{6} = \frac{3}{2}. \text{ Volume ratio} = k^3 = \frac{27}{8}.$$

- 22** Two similar spheres have surface areas in the ratio 9 : 25. Find the ratio of their volumes.
 Area ratio $= k^2 = \frac{9}{25} \Rightarrow k = \frac{3}{5}$. Volume ratio $= k^3 = \frac{27}{125}$.
- 23** A scale model of a statue is built at a 1 : 20 scale. The actual statue has volume 4000 m³. Find the volume of the model.
 Volume ratio $= \left(\frac{1}{20}\right)^3 = \frac{1}{8000}$. Model volume $= \frac{4000}{8000} = 0.5$ m³.
- 24** Two similar rectangular boxes have volumes in the ratio 8 : 27. Find the ratio of their surface areas.
 Volume ratio $= k^3 = \frac{8}{27} \Rightarrow k = \frac{2}{3}$. Surface area ratio $= k^2 = \frac{4}{9}$.

Cross-sections of solids

- 25** A cone has base radius 8. A plane parallel to the base slices the cone at half the height, measured from the apex. Find the radius of the circular cross-section at that height.
 The cross-section radius scales linearly with distance from the apex: radius $= 8 \times \frac{1}{2} = 4$.
- 26** A cube of side 12 is sliced by a plane through the midpoints of the three edges meeting at one vertex. Find the area of the resulting triangular cross-section.
 Each cut edge has length 6 (half of 12), and the two cut edges at each pair meet at a right angle, so each side of the triangular cross-section has length $\sqrt{6^2 + 6^2} = 6\sqrt{2}$. By symmetry the triangle is equilateral, so its area is $\frac{\sqrt{3}}{4}(6\sqrt{2})^2 = \frac{\sqrt{3}}{4}(72) = 18\sqrt{3}$.

A well-designed word problem: a grain silo

- 27** This question has three parts. A grain silo is shaped like a cylinder topped with a cone, both with radius 5 m. The cylindrical part has height 12 m, and the conical roof has height 3 m. Use $\pi \approx 3.14159$. (a) Find the total volume of the silo (cylinder plus cone), to the nearest cubic meter. (b) The silo is filled with grain up to the top of the cylindrical part only. If each cubic meter of grain has a mass of 750 kg, find the total mass of grain stored, to the nearest kilogram. (c) A second, similar silo is built at 1.4 times the linear scale of the first, in every dimension. Find the volume of the second silo, to the nearest cubic meter.
 (a) Cylinder: $\pi(25)(12) = 300\pi \approx 942.48$ m³. Cone: $\frac{1}{3}\pi(25)(3) = 25\pi \approx 78.54$ m³. Total ≈ 1021.02 , so 1021 m³. (b) Cylinder volume ≈ 942.48 m³. Mass $\approx 942.48 \times 750 \approx 706858$ kg. (c) Volume scales as $k^3 = 1.4^3 = 2.744$. New volume $\approx 1021.02 \times 2.744 \approx 2802$ m³.